

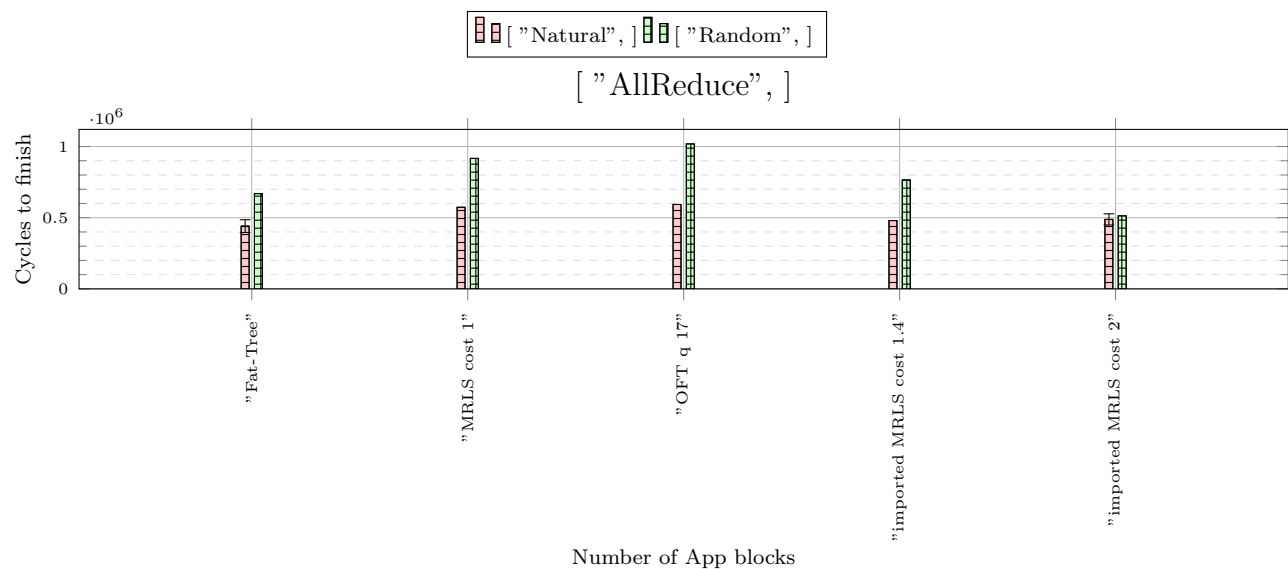


Figure 1: X: ["AllReduce",]


```

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    ArgumentVC{
      allowed: [ 2 ]},
    ArgumentVC{
      allowed: [ 2 ]},
    ArgumentVC{
      allowed: [ 3 ]},
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      allowed: [ 3 ]}],
  VecLabel{
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  OccupancyFunction{ label_coefficient: 1, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 0, use_internal_space: true,
use_neighbour_space: true, aggregate: true },
  OccupancyFunction{ label_coefficient: 1, occupancy_coefficient: 1, product_coefficient: 0, constant_coefficient: 0, use_internal_space: true,
use_neighbour_space: true, aggregate: false },
  LowestLabel,
  EnforceFlowControl,
  Random],
[
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use_neighbour_space: true, aggregate: true },
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  LowestLabel,
  EnforceFlowControl,
  Random],
[
  MapHop{
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      ArgumentVC{
        allowed: [ 1 ]},
      ArgumentVC{
        allowed: [ 1 ]},
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        allowed: [ 2 ]},
      ArgumentVC{
        allowed: [ 2 ]},
      ArgumentVC{
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use_neighbour_space: true, aggregate: false },
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    EnforceFlowControl,
    Random]],
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  flit_size: 16,
  intransit_priority: false,
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  output_buffer_size: 64,
  crossbar_frequency_divisor: 1,
  crossbar_delay: 2},
routing: topo![
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  UpDown{ legend_name: "minimal routing" },
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